

PathFinder — Testing Documentation

1. Testing Strategy

PathFinder employs a multi-layered testing strategy focused on ensuring the reliability and correctness of all core features across the full stack.

Testing Levels

- Manual Functional Testing:** Primary testing method for the MVP. Each feature module is tested end-to-end through the UI and API.
- API Endpoint Testing:** Direct HTTP requests to backend endpoints to verify correct responses, error handling, and authorization.
- Build Verification:** TypeScript compilation and Vite build processes catch type errors and import issues.
- Deployment Validation:** Post-deployment checks on Vercel and Render to verify the live environment matches local behavior.

2. Test Cases — Authentication Module

ID	Test Case	Steps	Expected Result	Status
AUTH-01	Student Registration	Register with valid email, password, role=STUDENT	201, token returned	☒ Pass
AUTH-02	Organization Registration	Register with valid email, password, role=ORGANIZATION, companyName	201, token returned	☒ Pass
AUTH-03	Duplicate Email Registration	Register with an already-used email	400, "Email already exists"	☒ Pass
AUTH-04	Valid Login	Login with correct credentials	200, token + user object	☒ Pass
AUTH-05	Invalid Password	Login with wrong password	400, "Invalid credentials"	☒ Pass
AUTH-06	Non-existent Email Login	Login with unregistered email	400, "Invalid credentials"	☒ Pass
AUTH-07	Rate Limiting	Send 6 login requests within 15 minutes	429 on 6th request	☒ Pass
AUTH-08	Forgot Password	Request reset for valid email	200, reset token saved to DB	☒ Pass
AUTH-09	Reset Password	Use valid token to set new password	200, password updated	☒ Pass
AUTH-10	Expired Reset Token	Use token after 1 hour	400, "Invalid or expired"	☒ Pass

3. Test Cases — Application Module

ID	Test Case	Expected Result	Status
APP-01	Submit application with cover letter	201, application created with PENDING status	☒ Pass
APP-02	Submit duplicate application	400, "Already applied"	☒ Pass
APP-03	Withdraw pending application	200, status changed to WITHDRAWN	☒ Pass
APP-04	Withdraw accepted application	400, "Cannot withdraw accepted"	☒ Pass
APP-05	Org updates status to ACCEPTED	200, student receives notification	☒ Pass
APP-06	Batch status update	200, all selected applications updated	☒ Pass
APP-07	AI match score generation	Match score (0-100) appears on application after submission	☒ Pass
APP-08	View application timeline	Timeline shows SUBMITTED event	☒ Pass

4. Test Cases — AI Module

ID	Test Case	Expected Result	Status
AI-01	Resume PDF parsing (with API key)	JSON profile data extracted from PDF	☒ Pass
AI-02	Resume PDF parsing (mock mode)	Mock profile data returned	☒ Pass
AI-03	Generate SOP	Full SOP text with student name and opportunity	☒ Pass
AI-04	Generate Cover Letter	Professional cover letter text	☒ Pass

AI-05 Generate Essay	Essay with custom topic	☒ Pass
AI-06 Profile Strength (100% profile)	Score = 100, all checks pass	☒ Pass
AI-07 Profile Strength (empty profile)	Score < 50, failed checks listed	☒ Pass
AI-08 Recommendations with matching skills	Top results match student skills	☒ Pass

5. Test Cases — Authorization

ID	Test Case	Expected Result	Status
AUTHZ-01	Student accesses org endpoint	403 Forbidden	☒ Pass
AUTHZ-02	Org accesses student endpoint	403 Forbidden	☒ Pass
AUTHZ-03	Non-admin accesses admin endpoint	403 Forbidden	☒ Pass
AUTHZ-04	Request without token	401 Unauthorized	☒ Pass
AUTHZ-05	Request with expired token	401 Unauthorized	☒ Pass
AUTHZ-06	Student views another's ticket	403 Access Denied	☒ Pass
AUTHZ-07	Admin deletes own account	400 "Cannot delete self"	☒ Pass

6. Test Cases — Support Ticketing

ID	Test Case	Expected Result	Status
TKT-01	Create ticket	201, ticket created with OPEN status	☒ Pass
TKT-02	View own tickets	200, returns user's tickets only	☒ Pass
TKT-03	Admin views all tickets	200, returns all tickets	☒ Pass
TKT-04	Add message to ticket	201, message appears in thread	☒ Pass
TKT-05	Non-creator views ticket	403 Access Denied	☒ Pass
TKT-06	Admin updates status	200, status changed to RESOLVED	☒ Pass

7. Build Verification

Check	Command	Result
Backend TypeScript compilation	<code>npx tsc --noEmit</code>	Compiles with 0 errors
Frontend Vite build	<code>vite build</code>	Build successful, dist/ generated
Prisma client generation	<code>npx prisma generate</code>	Client generated successfully
Database schema push	<code>npx prisma db push</code>	Schema applied to SQLite

8. Known Issues

Issue	Severity	Status	Workaround
Render free tier cold start (50s)	Low	Known	Users informed in documentation
TypeScript strict mode warnings (unused imports)	Low	Accepted	Vercel build command overridden to <code>vite build</code>
SQLite concurrent write limitations	Low	Accepted	Suitable for current user scale

End of Testing Documentation